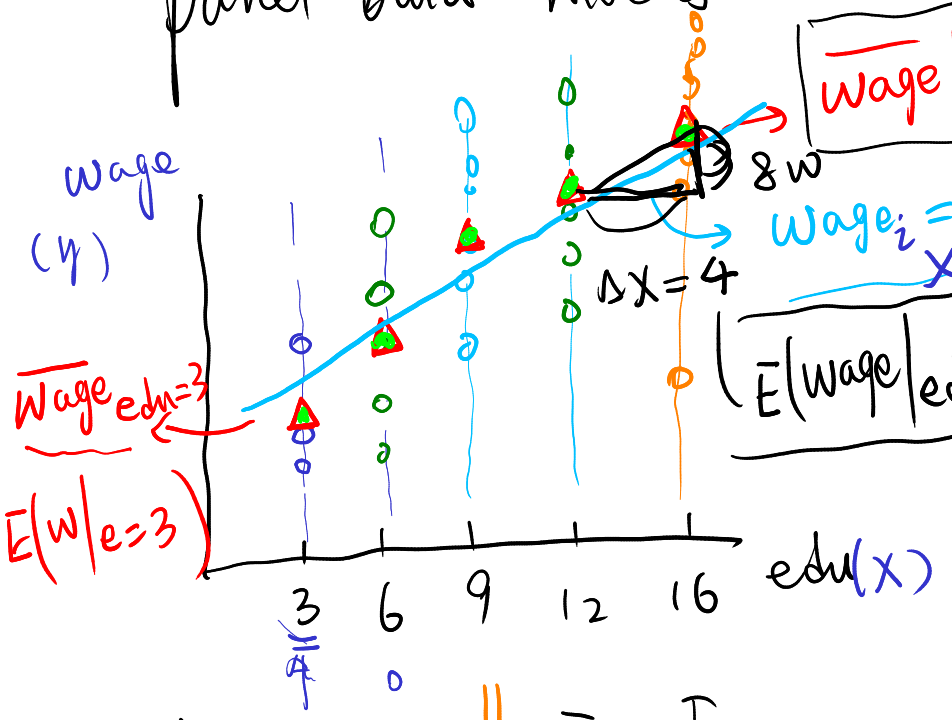
Panel Data Models

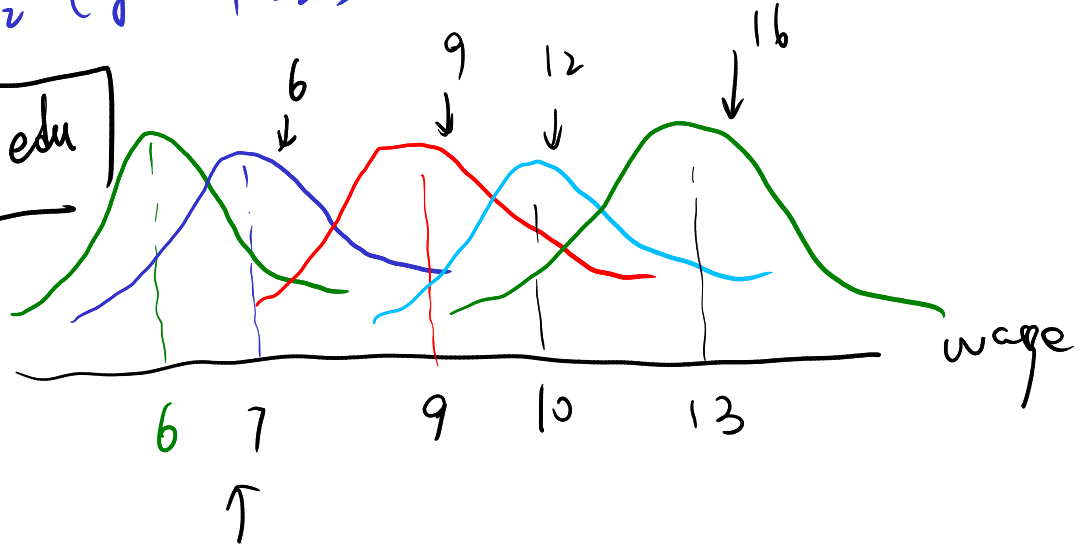


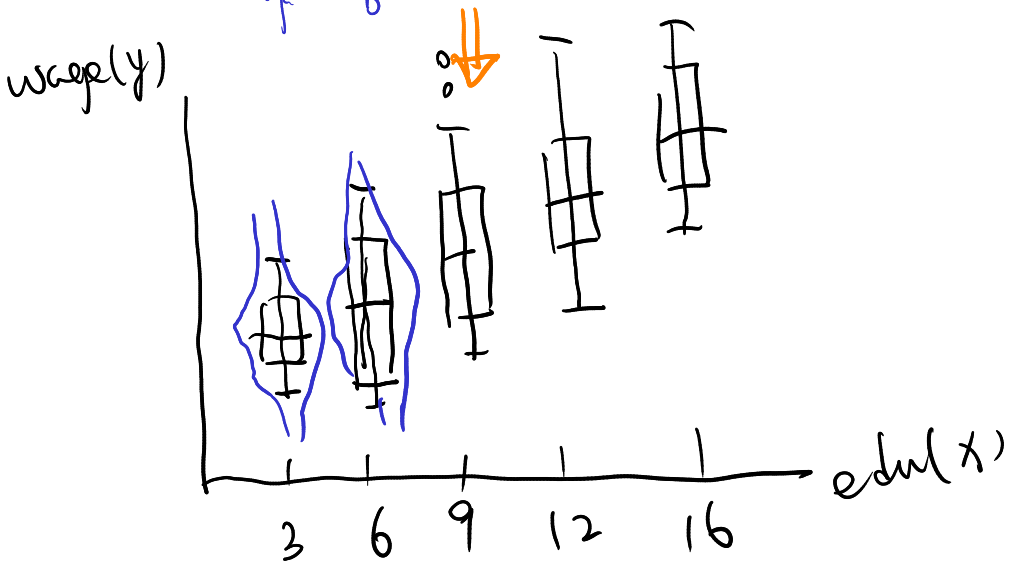
$\boxed{\overline{wage}\,|\,edu=16} \Longleftrightarrow E(wage\,|\,edu=16)$ (条件期望)

$wage_i = a + b \cdot edu_i$ $(y = a + bx)$

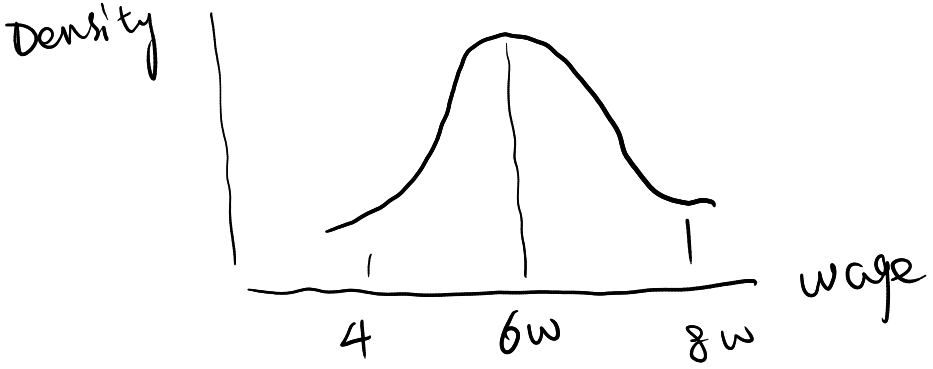
$\boxed{E(wage|edu) = a + \beta \cdot edu}$

(条件 model)

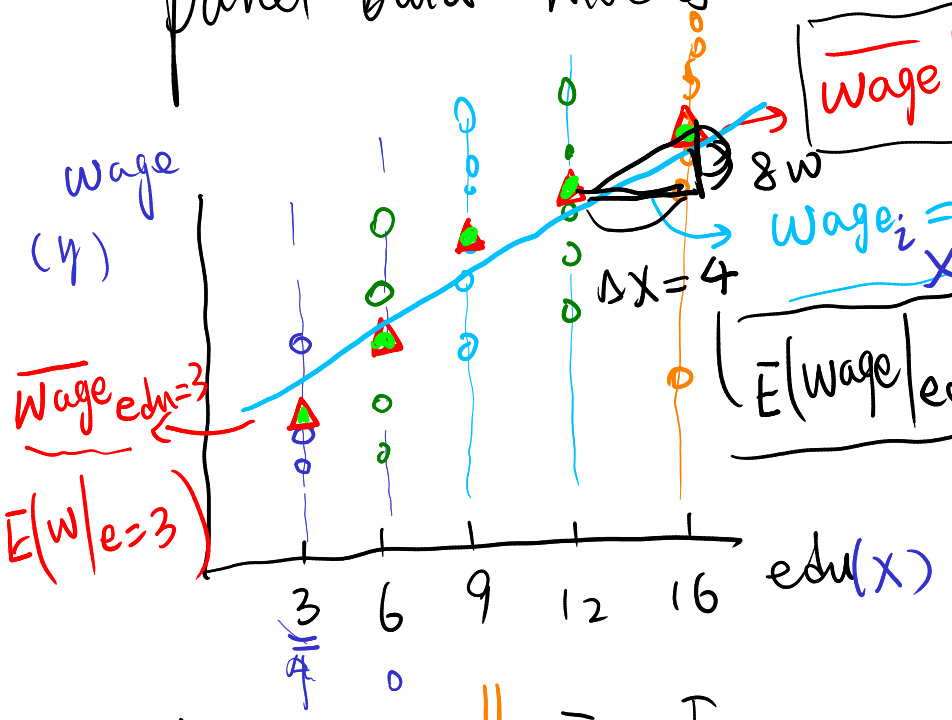




kdensity wage if edu=6 (条)



Panel Data Models

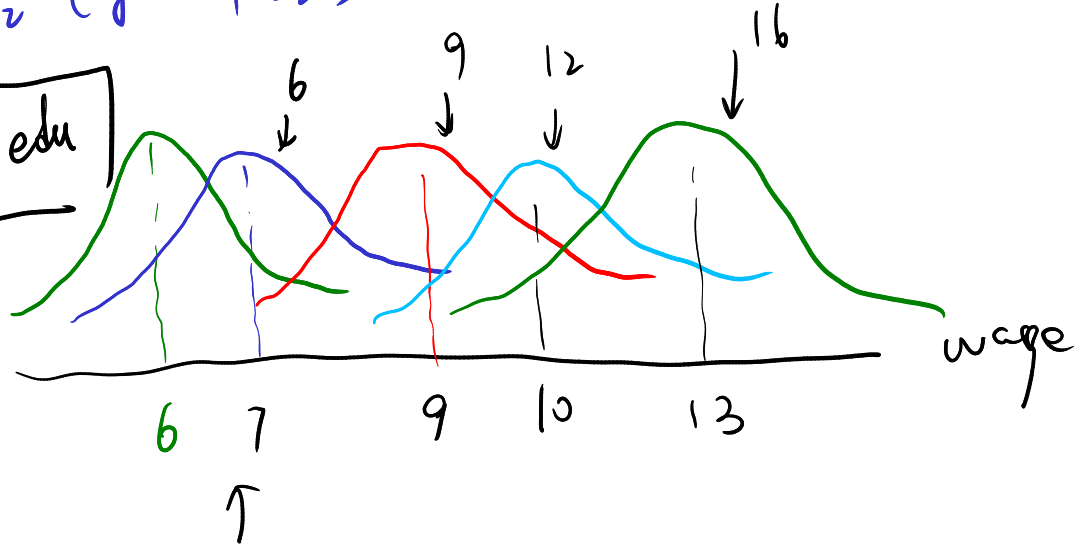


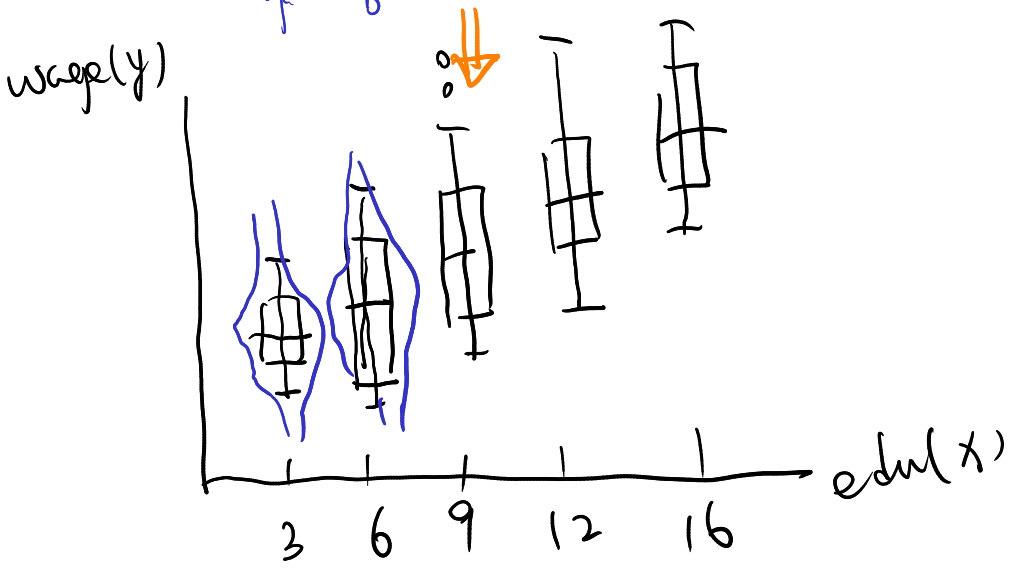
$\boxed{\overline{wage}\,|\,edu=16} \Longleftrightarrow E(wage\,|\,edu=16)$ (条件期望)

$wage_i = a + b \cdot edu_i$ $(y = a + bx)$

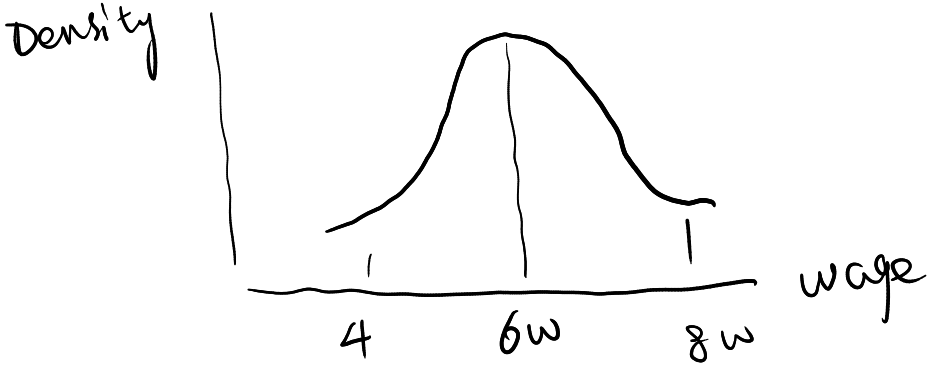
$\boxed{E(wage|edu) = a + \beta \cdot edu}$

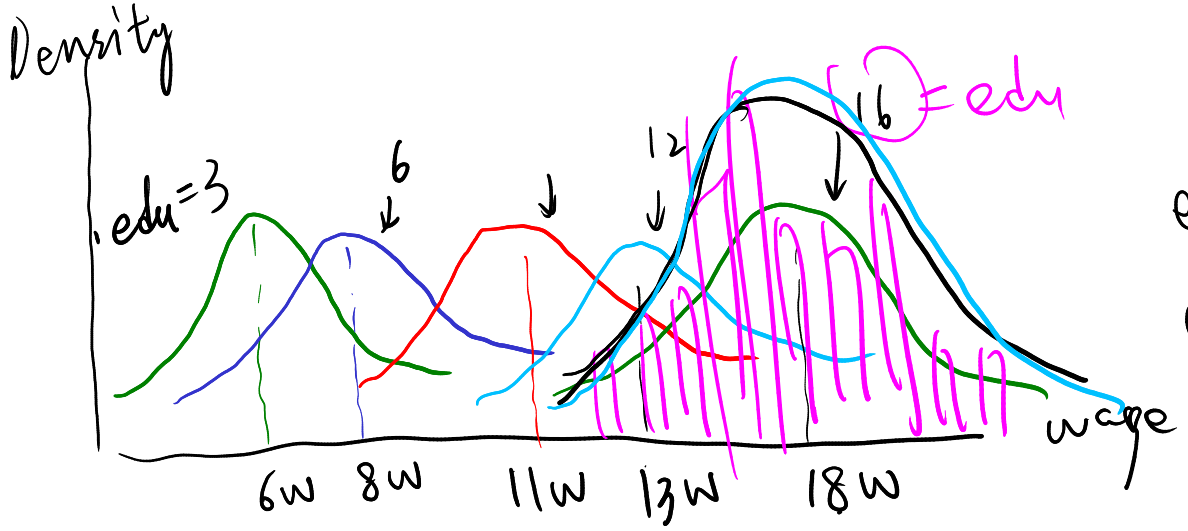
(条件 model)

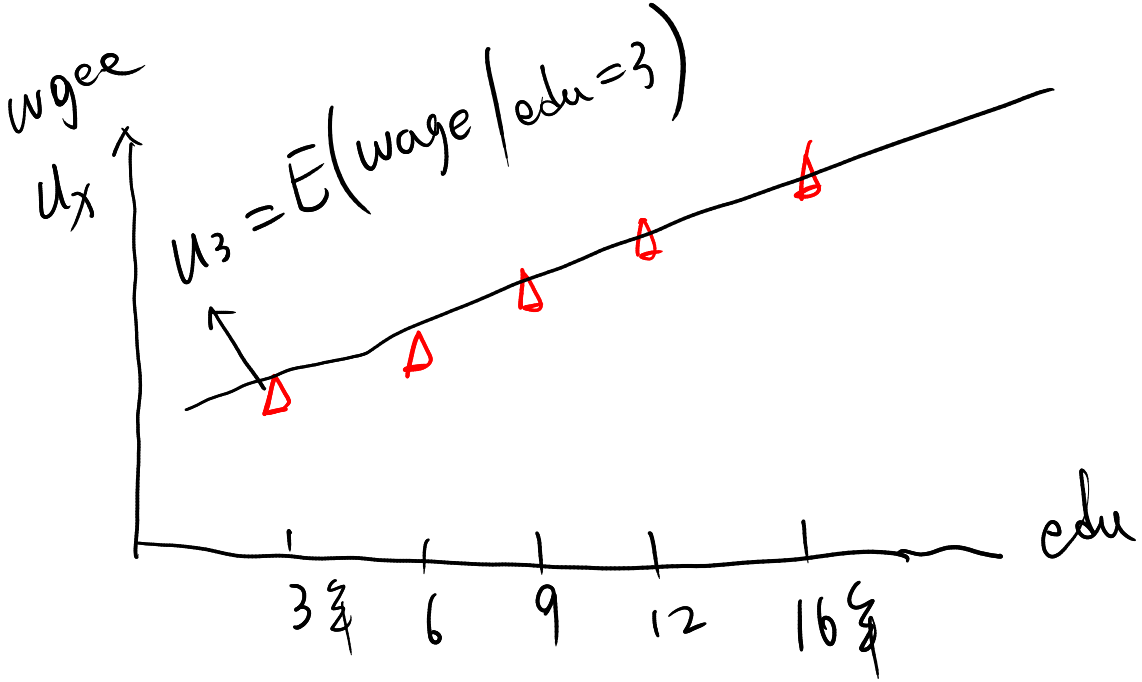




kdensity wage if edu=6 (条)







MLE :

edu = 3 : $\text{wage} \sim N(u_3, \sigma_3^2)$

edu = 6 : $\text{wage} \sim N(u_6, \sigma_6^2)$

$$\boxed{\text{wage} \sim N(u_{(x)}, \sigma^2_{(x)})}$$

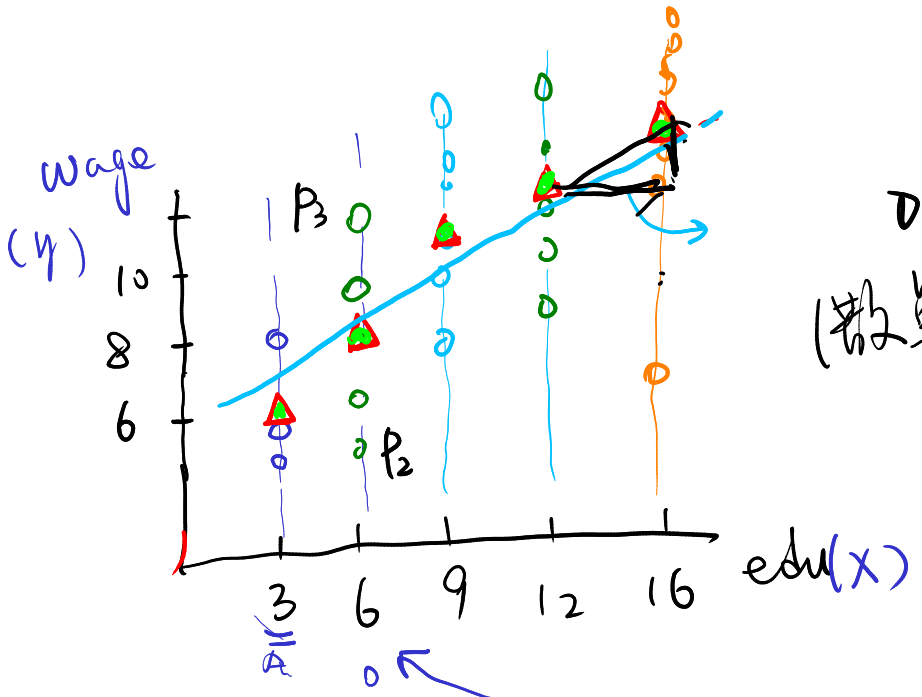
$E(\text{wage} \mid \text{Edu})$

参数

降维

$u_x = \alpha + \beta \cdot \text{edu}$

$\sigma_x^2 = \exp(\theta_1 + \theta_2 \cdot \text{edu})$



Model: ① $E(y|x) = \alpha + \beta \cdot x$ CEF Conditional Expectation Function

data ↓ ② $y_i = \alpha + \beta \cdot x_i + \boxed{\varepsilon_i}$ residuals 残差项

(散点)

$i = 1, 2, \sim N$ $(N = 100人)$

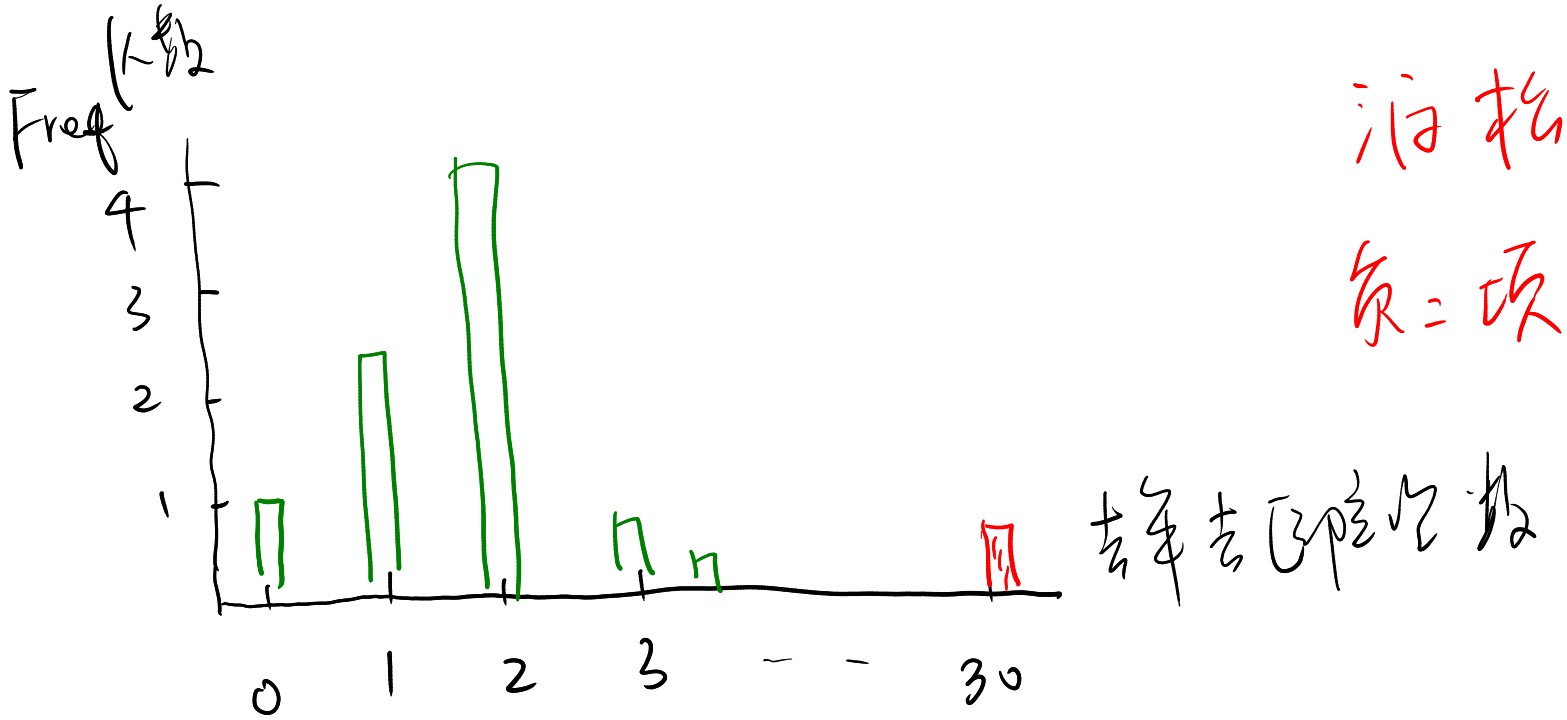
$E(\varepsilon|x) = 0$

$E(\varepsilon|edu=6) = \sum_{i=1}^{N_6} \varepsilon_i = 0$

$wage_i = \alpha + edu_i \cdot \beta_1 + Ability_i \cdot \beta_2 + Sex_i \cdot \beta_3 + \boxed{\cdots}$

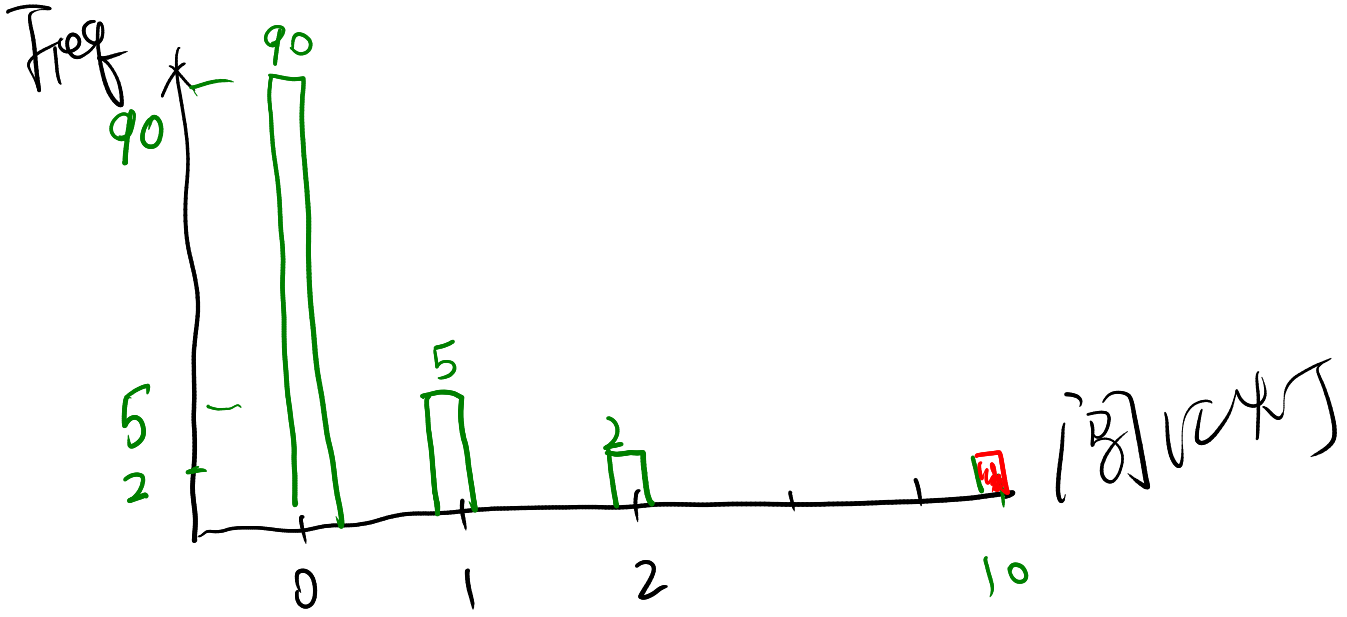
对②式子…可以 证明：

$E(y_i|x_i) = \alpha + \beta x_i + \boxed{E(\varepsilon_i|x_i)} = 0$



泊松 poisson

负二项 NB



N=100人

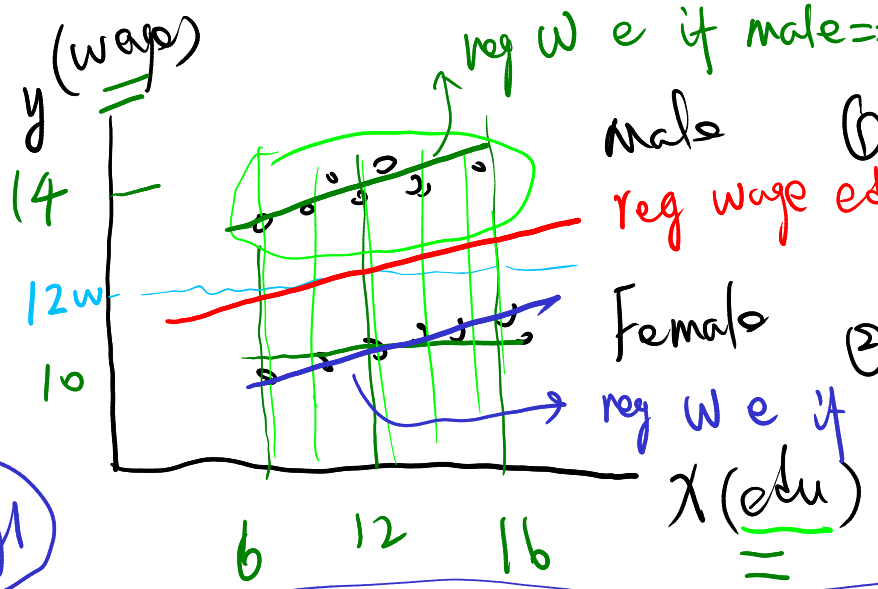


Fig1

Male ① $wage_i = a + \beta_1 \cdot edu + \varepsilon_i$ → $Sex_i \cdot \beta_2$

Female ② $Wage_i = a + \beta_1 \cdot edu_i + \beta_2 \cdot Sex_i + u_i$?

$corr(edu, sex) \overset{?}{=} 0$

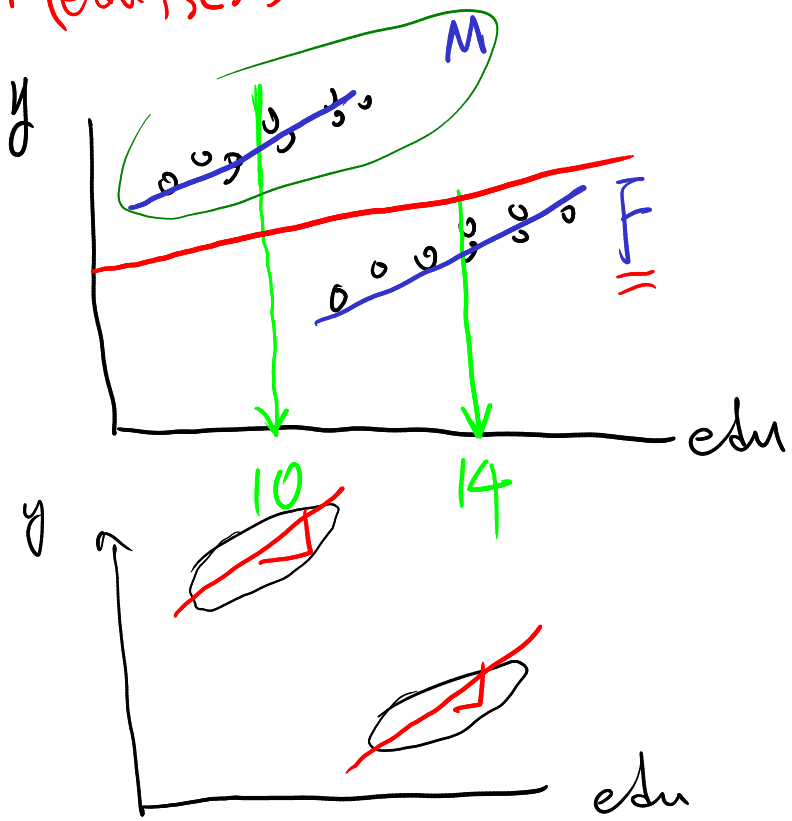
目标? A, $\hat{wage_i} \mid edu = ?$ 预测

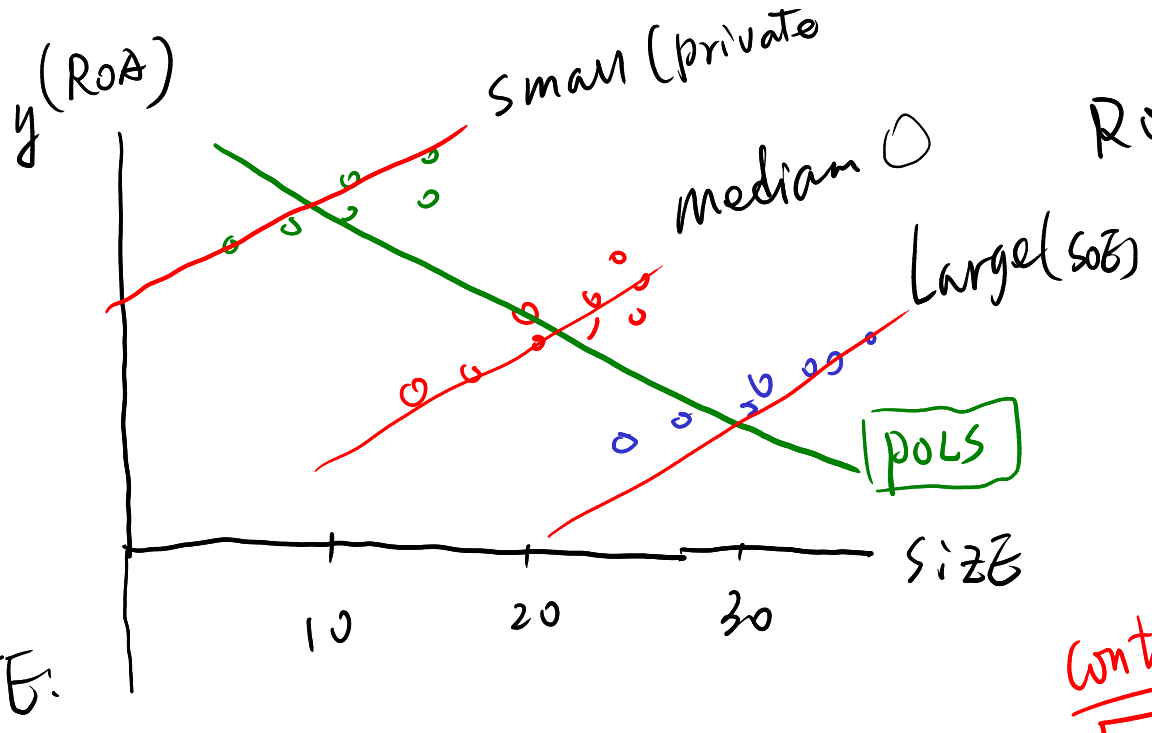
B, $\frac{\Delta Wage}{\Delta edu} = ?$

(x) edu → wage (y)

sex → edu, sex → wage

Fig2





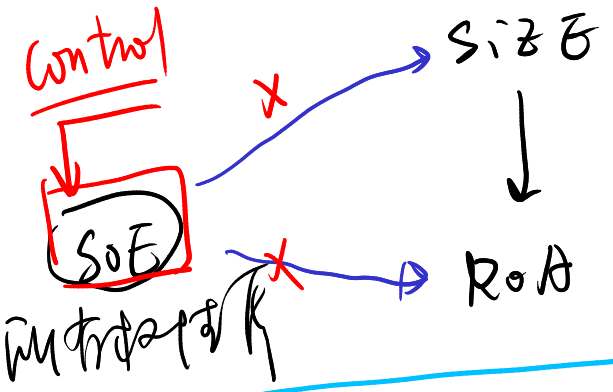
$$ROA_{it} = a + size_{it}\beta_1 + \varepsilon_{it}$$

POLS

- reg ROA size

  $\hat{\beta} = -0.3^{***}$

- reg ROA size SOE



FE:

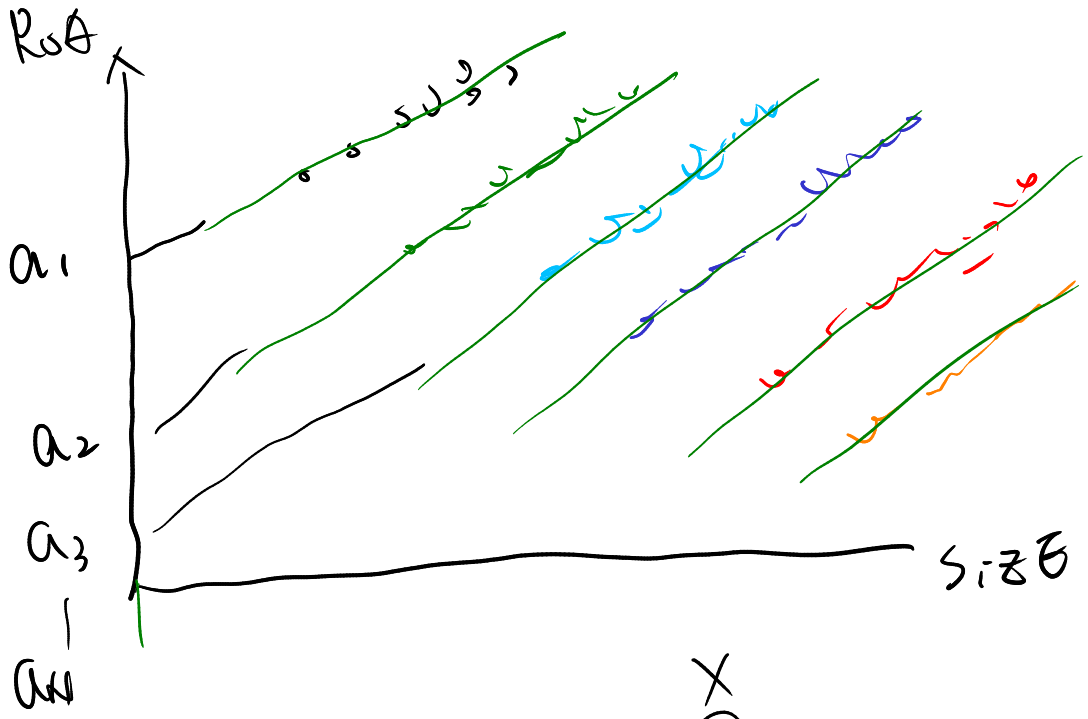
$$y_{it} = a_i + x_{it}\beta + \varepsilon_{it}$$

- reg y x iid
- reghdfe y x, absorb(id)

TWFE

$$y_{it} = a_i + \lambda_t + x_{it}\beta + \varepsilon_{it}$$

reghdfe y x, absorb(id year)



$$RoA_{it} = \boxed{X_{it}}\cdot\beta_1 + W_i\beta_2 + W_t\cdot\beta_3 + \varepsilon_{it}$$

- X_{it}: $Size_{it}$, $Leverge_{it}$, $cflow_{it}$ — ?
- W_i: ✓ 行业特征, ? CEO特征 — ?
- W_t: 宏观 M_{2t} ✓, ue_t ✓ — ?

不可观测的因素

$$y_{it} = X_{it}\beta + a_i + \lambda_t + \varepsilon_{it}$$

$i=1$: $y_{1t} = X_{1t}\beta + a_1\cdot D_1 + \varepsilon_{1t}$, $E(y_{1t} \mid X_{1t}=0) = a_1$

$i=2$: $y_{2t} = X_{2t}\beta + a_2\cdot D_2 + \varepsilon_{2t}$, $\cancel{E(y_{2t} \mid X_{2t}=0)} = a_2$

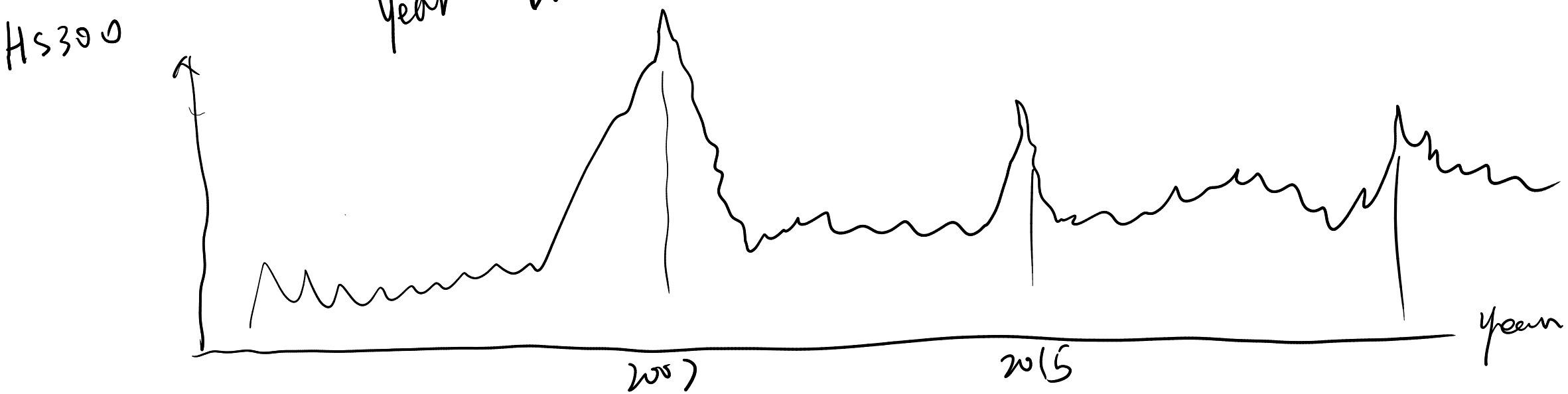
$$y_{it} = X_{it}\beta + \sum_{i=1}^{N} a_i\cdot \boxed{D_i} + \varepsilon_{it}$$

所有不随时间变化的因素都包含.

$$y_{it} = \underbrace{a_i}_{\text{FE}} + X_{it}\beta + \varepsilon_{it}$$

TWFE

$$y_{it} = a_i + X_{it}\beta + \underbrace{D_t}_{\text{FE (year)}} + \varepsilon_{it}$$

$$D_t = \sum_{t=1}^{T} \lambda_t \cdot \underbrace{M_t}_{\text{Year Dummies}}$$

id	year	D_1	D_2	D_3	M_1	M_2	M_3
1	2021	1	0	0	1	0	0
1	2022	1	0	0	0	1	0
1	2023	1	0	0	0	0	1
2	2021	0	1	0	1	0	0
2	2022	0	1	0	0	1	0
2	2023	0	1	0	0	0	1
3	2021	0	0	1	1	0	0
3	22	0	0	1	0	1	0
3	23	0	0	1	0	0	1